

LAB #4: SLAM AND NAV2 (ROS 2 HUMBLE)

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I. SLAM AND NAV2 WITH my_robot

This lab guides you through simulation, mapping (SLAM Toolbox), and localization/navigation (Nav2) using my_robot.

The following prerequisites are required:

- ROS 2 Humble installed and sourced
- Gazebo and RViz2 installed
- my_robot built and sourced

```
colcon build && source install/setup.bash
```

- Robot publishes /robot_description and a laser /scan

A. Simulation + RViz

Displays the robot in Gazebo and publishes /robot_description and /scan for RViz.

```
ros2 launch my_robot launch_sim.launch.py
```

In RViz:

- Set Fixed Frame (e.g., odom)
- Add RobotModel (uses /robot_description)
- Add LaserScan on /scan

B. Mapping with SLAM Toolbox

Start online SLAM and drive the robot to cover the space.

```
ros2 launch my_robot launch_slam.launch.py
```

Tip

- Teleop: `ros2 run teleop_twist_keyboard teleop_twist_keyboard`
- Check topics/frames: `ros2 topic list`, `ros2 run tf2_tools view_frames`
- View map in RViz: add Map (topic /map)

Save the map:

```
ros2 run nav2_map_server map_saver_cli -f ~/maps/my_world
```

C. Localization and Navigation (Nav2)

Launch Nav2 (AMCL + planners + controllers) and navigate in RViz.

```
ros2 launch my_robot launch_nav.launch.py
```

The overall workflow is as follows:

- In RViz set Fixed Frame to map
- Use 2D Pose Estimate to set initial pose
- Use Nav Goal to send a goal

D. Summary

```
# Simulation
ros2 launch my_robot launch_sim.launch.py

# Mapping
ros2 launch my_robot launch_slam.launch.py

# Localization + Navigation
ros2 launch my_robot launch_nav.launch.py
```